

Harrison Tidy

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EDUCATION

University of British Columbia

Bachelor of Applied Science - Computer Engineering | GPA: 4.0/4.0

Vancouver, BC

Expected Apr 2029

WORK EXPERIENCE

WriteAway AI | Founder & Head of Engineering

2024 - Present

- Founded WriteAway to build an AI study pen; hired engineers and lead a 4-person team developing the product.
- Oversaw miniaturized PCB design, component selection, and embedded C firmware integration across I2C, SPI, and DVP interfaces.
- Led rapid prototyping across nearly 100 mechanical prints, 5+ PCB revisions, and optical iterations; debugged hardware, firmware, and mechanical integration with the team.

Creation Technologies

Jun 2026 - Aug 2026

PCB Assembly & Soldering Technician

Burnaby, BC

- Assembled, hand-soldered, and inspected PCB assemblies for military, industrial, and medical applications while following ESD, workmanship, and production quality procedures.
- Used microscopes and production inspection equipment to identify soldering, component-placement, and assembly defects.
- Received internal IPC-A-610 and J-STD-001 training and applied the standards to PCB assembly and inspection.

TECHNICAL PROJECTS

AutoBOM | C++, Node.js, KiCad

2026

- Extended KiCad's C++/wxWidgets interface and connected a Node.js backend over HTTP/JSON to build an AI assistant for component sourcing and application-circuit generation.
- Automated DigiKey and JLCPCB/LCSC searches, symbol/footprint imports, and missing BOM-field completion with reviewed schematic placement.
- Benchmarked 50 component searches at a 15.8-second median and five application circuits at a 17.4-second median without reusing saved circuit recipes.

UBC Course Scheduler | React, TypeScript, Tailwind CSS

2026

- Built a course scheduling app with React, TypeScript, and Tailwind CSS, using React hooks for calendar views and term selection.
- Implemented recurring and alternate-week scheduling, date filtering, and duplicate meeting detection.

PickleBot | Autonomous Pickleball-Playing Robot

2026

- Developing an autonomous robot to play pickleball against, with Python/OpenCV, YOLO, and Roboflow ball/court tracking on Raspberry Pi 5 with a Hailo-8L AI accelerator.
- Integrating ESP32 embedded control over UART, PWM motor/servo control, and Bluetooth manual driving.
- Designed a KiCad power/control PCB and custom brackets; assembled actuators, shaft couplers, aluminum extrusion, and 3D-printed parts into the robot.

Bluetooth RC Car | ESP32, Bluetooth, PCB Design

2026

- Designed, assembled, and hand-soldered a custom ESP32 PCB integrating Bluetooth control, buzzers, LEDs, and H-bridges for an RC car.

TECHNICAL SKILLS

Software: C, C++, Python, Java, Git/GitHub, OpenCV, YOLO, Roboflow, AI-assisted coding

Embedded Systems: Embedded C, ESP32, ESP-IDF, FreeRTOS, Wi-Fi, BLE, UART, I2C, SPI, DVP, PWM

Hardware & Prototyping: KiCad, Fusion 360, soldering, PCB assembly, optical prototyping, FDM/resin 3D printing